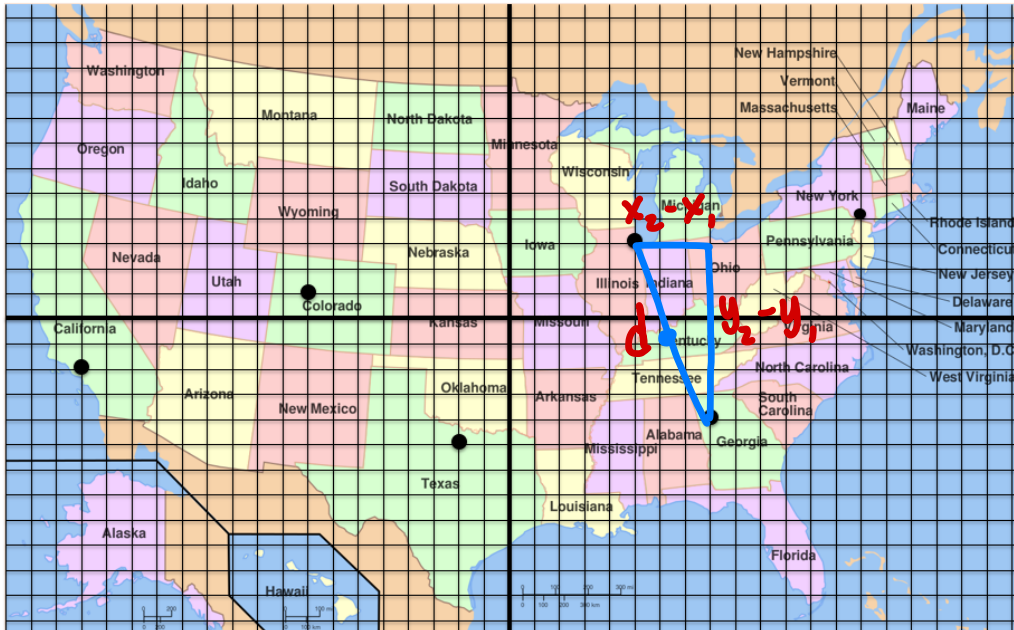


Are We There Yet?



According to the National Air Traffic Controllers Association there are over 87,000 flights that crisscross the United States every day! The top six busiest airports are labeled in the map below (Hartsfield-Jackson Atlanta, Los Angeles, Chicago O'Hare, Dallas, Denver, and John F. Kennedy in New York). Each unit represents approximately 75 miles.



1. A flight is departing Chicago and landing in Atlanta. Give ordered pairs for both airports.

Chicago = $(5, 3)$ Atlanta = $(8, -4)$

2. a. What is the horizontal distance between the two cities? Show this on the map.

3 units or 225 miles $x_2 - x_1 = 8 - 5$

- b. What is the vertical distance between the two cities? Show this on the map.

7 units or 525 miles $y_2 - y_1 = 3 - (-4)$

- c. To the nearest tenth of a mile, how far does the plane travel when it flies to Atlanta directly? How do you know?

$225^2 + 525^2 = c^2$ $c \approx 571.2$ miles
Distance is the hypotenuse of the right triangle.

3. Plot a point on the map that represents the half-way mark between Chicago and Atlanta.

- a. What are the coordinates of this point?

$(6.5, -0.5)$ midpoint

- b. If we only had the ordered pairs of the airports without the map, could we still find the coordinates of the half-way mark? How?

yes; average the 2 x-coordinates
then average the 2 y-coordinates

4. The Denver airport is halfway between LA and a new airport that will be built. What are the coordinates of the new airport? How do you know?

$(1, 4)$ From LA to Denver is over 9 up 3,
then we must go over 9 up 3 again.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Denver is equidistant from L.A. and new airport.

Lesson 0.1 – The Cartesian Plane

QuickNotes

Distance is the hypotenuse of a right Δ
Use Pythagorean Theorem!
 $a^2 + b^2 = d^2$

$(x_2 - x_1)^2 + (y_2 - y_1)^2 = d^2$
 $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = d$

Midpoint is equidistant from both endpoints.
 $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
* Average the x and y coordinates separately

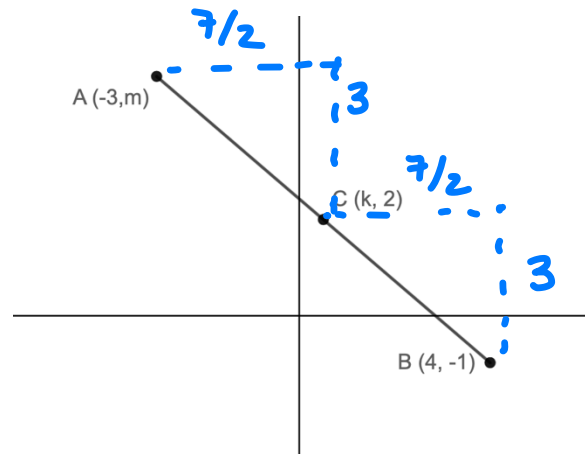
Check Your Understanding

- Kenya and Pete met up Starbucks and then drove home separately. Kenya drove 3 miles west and Pete drove 8 miles north. How far apart do Kenya and Pete live? Round to the nearest hundredth.

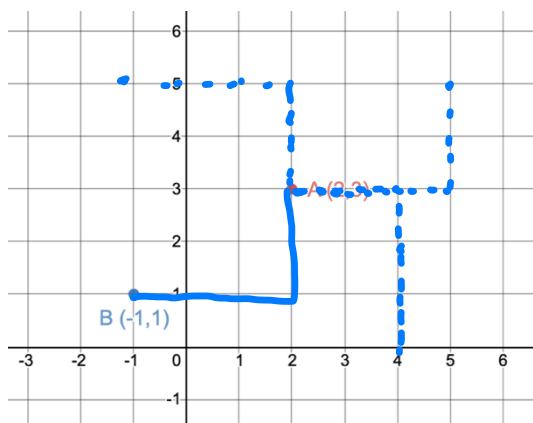
$3^2 + 8^2 = d^2$
 $73 = d^2$
 $d = \sqrt{73} \approx 8.54 \text{ miles}$

- C is the midpoint of segment $\overline{AB}$. Find the value of $m + k$.

$k = -3 + 3.5 = 0.5$
 $m = 2 + 3 = 5$
 $m + k = 5.5$



- Find three different locations for a new point C, such that $\overline{AB} \cong \overline{AC}$.



$C = (5, 5) \text{ or } (-1, 5) \text{ or } (4, 0)$
(Many answers possible)